# Docker 应用部署-Seata

参考链接：Docker部署 | Apache Seata

本示例以2.5.0版本为例，如果你要安装其他版本请自行修改版本号，可以通过下面链接查找镜像版本支持

- Seata Server：https://hub.docker.com/r/apache/seata-server/tags
- Naming Server：https://hub.docker.com/r/apache/seata-naming-server/tags

## 1 安装前准备

### 1.1 准备目录

创建一个专门的目录来存储seata相关配置文件和数据

```
mkdir -p /home/seata/2.5.0
cd /home/seata/2.5.0
```

### 1.2 拉取镜像

```
docker pull apache/seata-server:2.5.0.jdk17
docker pull apache/seata-naming-server:2.5.0.jdk17
```

### 1.3 运行镜像

```
docker run --name seata250 -d apache/seata-server:2.5.0.jdk17
```

### 1.4 检查结果

通过docker ps | grep seata命令查看，如果看到下图所示的效果表示运行成功。

```
[root@7c10c9d2eb8d 2.5.0]# docker ps | grep seata
6295641b86ae   apache/seata-server:2.5.0   "/bin/bash /seata-se…"   18 seconds ago   Up 17 seconds
[root@7c10c9d2eb8d 2.5.0]#
```

如果运行不成功，根据提示解决问题。多半是端口占用，或者容器名称重复等等。

### 1.5 获取Seata配置

从刚才创建好的容器中获取配置文件

```
docker cp seata250:/seata-server/resources /home/seata/2.5.0/seata-config
```

```
[root@7c10c9d2eb8d 2.5.0]# docker cp seata250:/seata-server/resources /home/seata/2.5.0/seata-config
Successfully copied 211kB to /home/seata/2.5.0/seata-config
[root@7c10c9d2eb8d 2.5.0]# ls seata-config/
application.example.yml  application.raft.example.yml  application.yml  banner.txt  docker  io  jdbc  logback  logback-spring.xml  lua
[root@7c10c9d2eb8d 2.5.0]#
```

## 1.6 获取Seata库

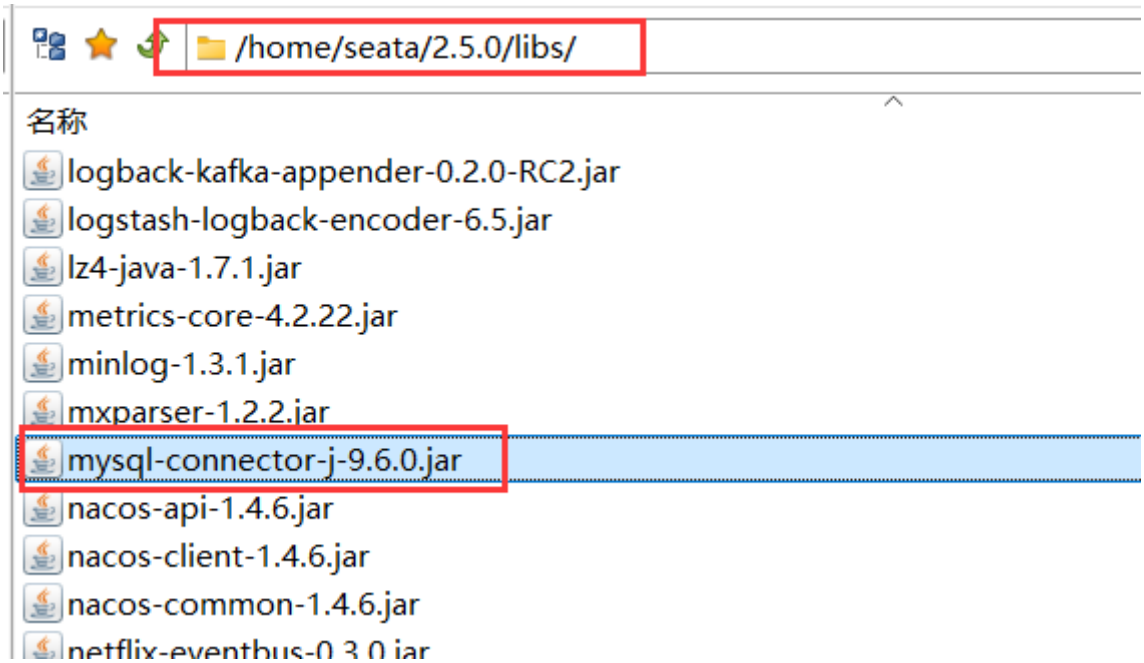
为了方便管理和调整seata的库文件，这里将他所有的库复制主机目录，作为数据卷目录提供。

```
docker cp seata250:/seata-server/libs /home/seata/2.5.0/libs
```

```
[root@7c10c9d2eb8d 2.5.0]# docker cp seata250:/seata-server/libs /home/seata/2.5.0/libs
Successfully copied 165MB to /home/seata/2.5.0/libs
[root@7c10c9d2eb8d 2.5.0]# ls libs/
ant-1.10.12.jar                    grpc-protobuf-1.55.1.jar          log4j-api-2.17.2.jar
ant-launcher-1.10.12.jar           grpc-protobuf-lite-1.55.1.jar     log4j-to-slf4j-2.17.2.jar
aopalliance-1.0.jar                grpc-stub-1.55.1.jar              logback-classic-1.2.12.jar
apollo-client-2.0.1.jar            gson-2.9.1.jar                    logback-core-1.2.12.jar
apollo-core-2.0.1.jar              guava-32.1.3-jre.jar              logback-kafka-appender-0.2.0-RC2.jar
archaius-core-0.7.6.jar            guice-5.0.1.jar                   logstash-logback-encoder-6.5.jar
asm-6.0.jar                        hamcrest-2.2.jar                  lz4-java-1.7.1.jar
audience-annotations-0.12.0.jar    hamcrest-core-2.2.jar             metrics-core-4.2.22.jar
bolt-1.6.7.jar                     hessian-4.0.3.jar                 minlog-1.3.1.jar
bucket4j_jdk8-core-8.1.0.jar       hessian-4.0.63.jar                mxparser-1.2.2.jar
checker-qual-3.37.0.jar            HikariCP-4.0.3.jar                nacos-api-1.4.6.jar
commons-codec-1.15.jar             httpasyncclient-4.1.5.jar         nacos-client-1.4.6.jar
commons-compiler-3.1.10.jar        httpclient-4.5.14.jar             nacos-common-1.4.6.jar
commons-configuration-1.10.jar     httpcore-4.4.16.jar               netflix-eventbus-0.3.0.jar
commons-dbcp2-2.9.0.jar            httpcore-nio-4.4.16.jar           netflix-infix-0.3.0.jar
commons-io-2.8.0.jar               j2objc-annotations-2.8.jar        netty-all-4.1.101.Final.jar
```

2.x 版本需要手动上传数据库驱动包到libs目录下面，不然你启动的时候会提示找不到驱动类。

从你的maven本地仓库中找一个对应版本MySQL驱动包，然后上传即可，上传后的效果如下。




## 1.7 停止与删除容器

```
docker stop seata250
docker rm seata250
```

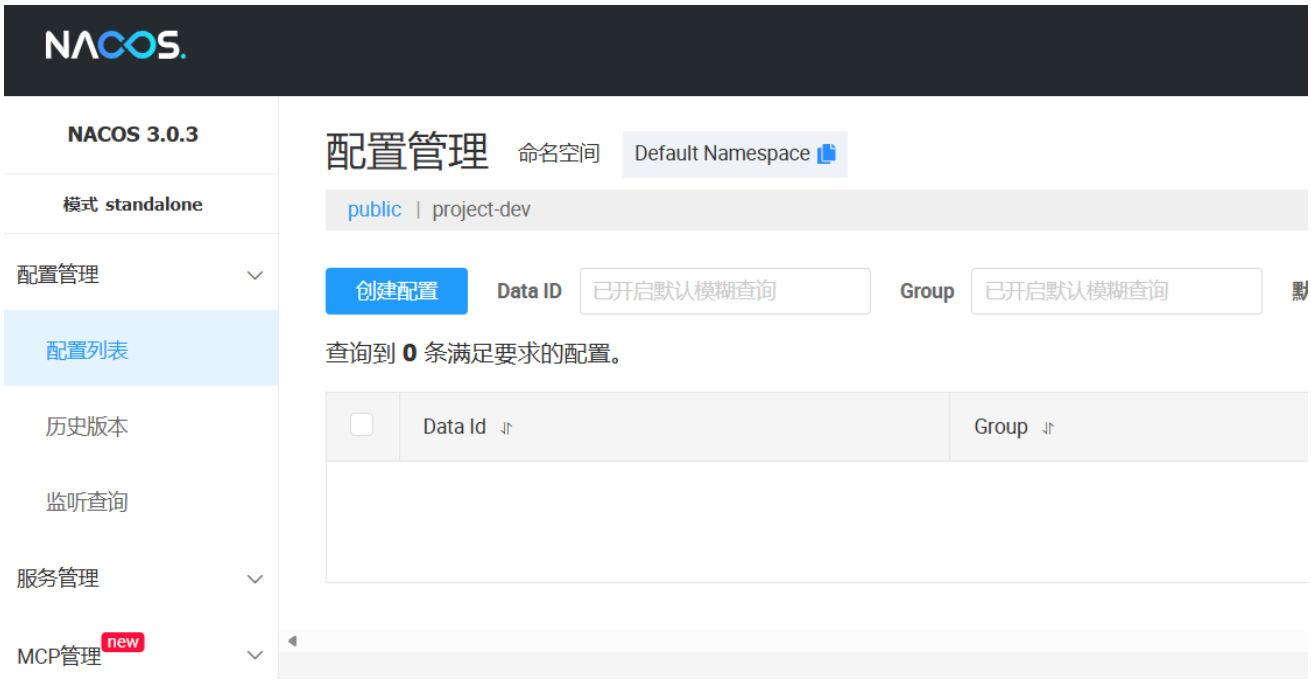
## 1.8 获取namingserver配置

获取这个配置和获取seata配置的方式是一样的，下面我就直接列举命令行即可

```
# 创建临时容器
docker run --name naming -d apache/seata-naming-server:2.5.0.jdk17
# 复制配置
docker cp naming:/seata-naming-server/resources /home/seata/2.5.0/naming-config
# 停止和删除容器
docker stop naming
docker rm naming
```

### 1.9 检查Nacos服务

如果没有启动nacos就参照之前的文档，启动后检查是否能在浏览器中访问nacos控制台




### 1.10 准备Seata事务库

执行资源/db/server-db.sql脚本即可。

### 1.11 配置防火墙

***注意：如果是云服务器端口放行需要到控制台页面去开放，一般情况操作的系统的防火墙是关闭的，不需要通过下面的命令行开放防火墙。***

```
firewall-cmd --add-port 8091/tcp --add-port 8092/tcp --permanent
firewall-cmd --reload
```

## 2 安装Seata

### 2.1 修改配置

修改seata-config/application.yml配置

```
vi seata-config/application.yml
```

配置内容参考如下，对应参考文件路径为资源/seata-config/application.yml，你可以直接上传到服务器覆盖对应文件后再进行修改。

```yaml
server:
  port: 8091
spring:
  application:
    name: seata-server
  main:
    web-application-type: none
logging:
  config: classpath:logback-spring.xml
  file:
    path: ${log.home:${user.home}/logs/seata}
  extend:
    logstash-appender:
      # off by default
      enabled: false
      destination: 127.0.0.1:4560
    kafka-appender:
      # off by default
      enabled: false
      bootstrap-servers: 127.0.0.1:9092
      topic: logback_to_logstash
      producer:
        acks: 0
        linger-ms: 1000
        max-block-ms: 0
    metric-appender:
      # off by default
      enabled: false
seata:
  config:
    type: nacos
    nacos:
      server-addr: 192.168.28.132:8848
      group: SEATA_GROUP
      namespace: ""
      cluster: default
      username: nacos
      password: nacos
      data-id: seataServer.properties
  registry:
    type: nacos,seata
    nacos:
      application: seata-server
      server-addr: 192.168.28.132:8848
      group: SEATA_GROUP
      namespace: ""
      cluster: default
      username: nacos
      password: nacos
    seata:
      server-addr: 192.168.28.132:8092
      cluster: default
      namespace: public
      heartbeat-period: 5000
```

```
      metadata-max-age-ms: 30000
      username: ${SEATA_CONSOLE_USERNAME:seata}
      password: ${SEATA_CONSOLE_PASSWORD:seata}
      tokenValidityInMilliseconds: 1740000
```

***注意：需要修改上面seata.config、seata.registry节点对应的IP、端口以及用户名和密码。***

详细配置可以参考application.example.yml，该文件存放了所有可使用的详细配置。

修改naming-config/application.yml

```
vi naming-config/application.yml
```

配置内容参考如下，对应参考文件路径为资源/naming-config/application.yml，你可以直接上传到服务器覆盖对应文件后再进行修改。

```
server:
  port: 8081
spring:
  application:
    name: seata-namingserver
logging:
  config: classpath:logback-spring.xml
  file:
    path: ${log.home:${user.home}/logs/seata}
heartbeat:
  threshold: 90000
  period: 60000
seata:
  security:
    secretKey: ${SEATA_SECRET_KEY:YSoD9q8eEALfqbe1q9KRlotpeoqqxisq2xZxurCIPtqBAda6N1R9OAbRm4Q3rJ71}
    tokenValidityInMilliseconds: 1800000
    csrf-ignore-urls: /naming/v1/**,/api/v1/naming/**
    ignore:
      urls: /,/**/*.css,/**/*.js,/**/*.html,/**/*.map,/**/*.svg,/**/*.png,/**/*.jpeg,/**/*.ico,/api/v1/auth/login,/version.json,/naming/v1/health,/error
console:
  user:
    username: ${SEATA_CONSOLE_USERNAME:seata}
    password: ${SEATA_CONSOLE_PASSWORD:seata}
```

## 2.2 服务编排

在/home/seata/2.5.0目录中建立docker-compose.yaml，并写入下列内容。

```
services:
  seata-naming:
    image: apache/seata-naming-server:2.5.0.jdk17
    networks:
      - seata
```

```
    hostname: seata-naming
    container_name: seata-naming
    ports:
      - "8092:8081"
    environment:
      - TZ=Asia/Shanghai
    volumes:
      - ./naming-config:/seata-naming-server/resources
      - ./naming-logs:/root/logs
  seata-server:
    image: apache/seata-server:2.5.0.jdk17
    networks:
      - seata
    hostname: seata-server
    container_name: seata-server
    ports:
      - "8091:8091"
    environment:
      - SEATA_PORT=8091
      - SEATA_IP=192.168.28.132
      - TZ=Asia/Shanghai
    volumes:
      - ./seata-config:/seata-server/resources
      - ./libs:/seata-server/libs
      - ./sessionStore:/seata-server/sessionStore
      - ./logs:/root/logs
    depends_on:
      - seata-naming
networks:
  seata:
    name: seata
    driver: bridge
```

注意：修改成你的服务器`IP`

## 2.3 启动服务

首先前台启动服务试试，执行下面的命令

```
docker compose up
```

启动成功后可以看到下图所示的效果




进入Nacos注册中心控制面板查看是否有Seata服务，如下图所示。

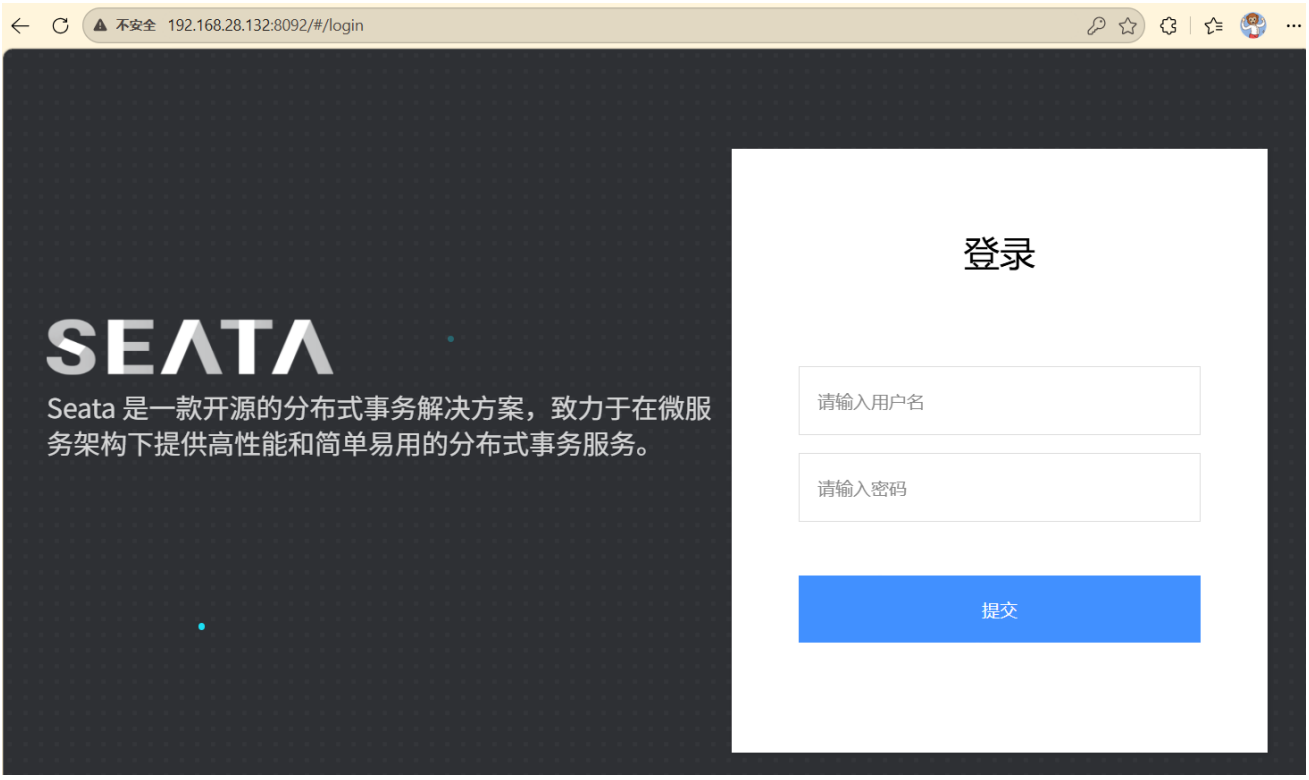



如果一切正常，那么服务器安装成功了，接下来可以切换到后台启动模式。

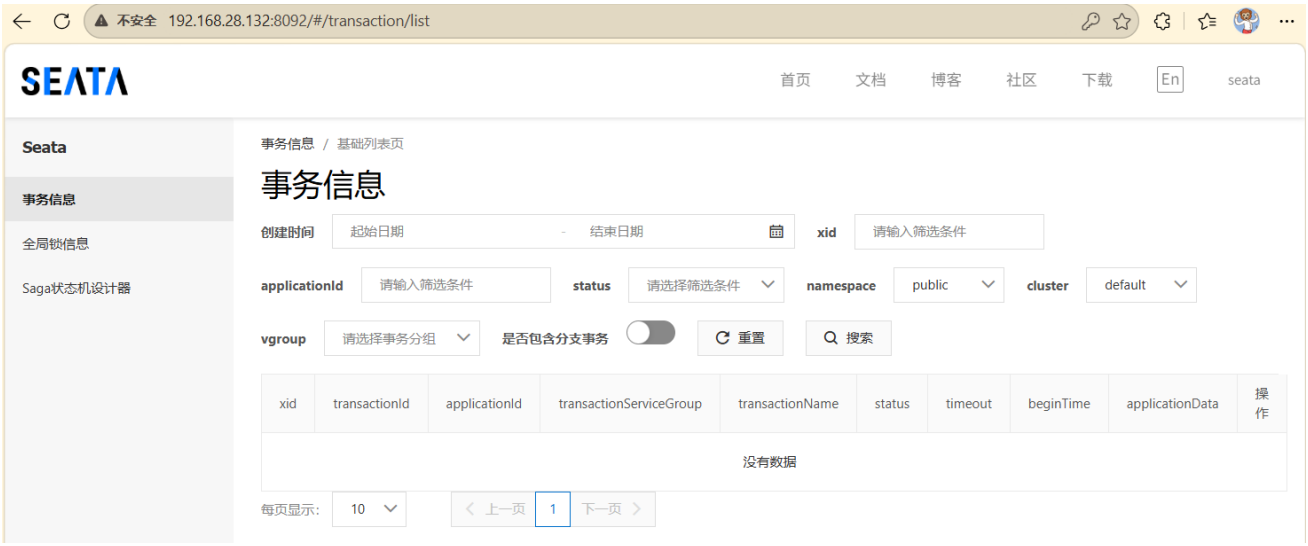
终端执行Ctrl + C结束前台启动，使用下面命令进行后台启动。

```
docker compose up -d
```

访问seata控制台，我的教程中控制台的端口是8092，所以访问地址是http://ip:8092



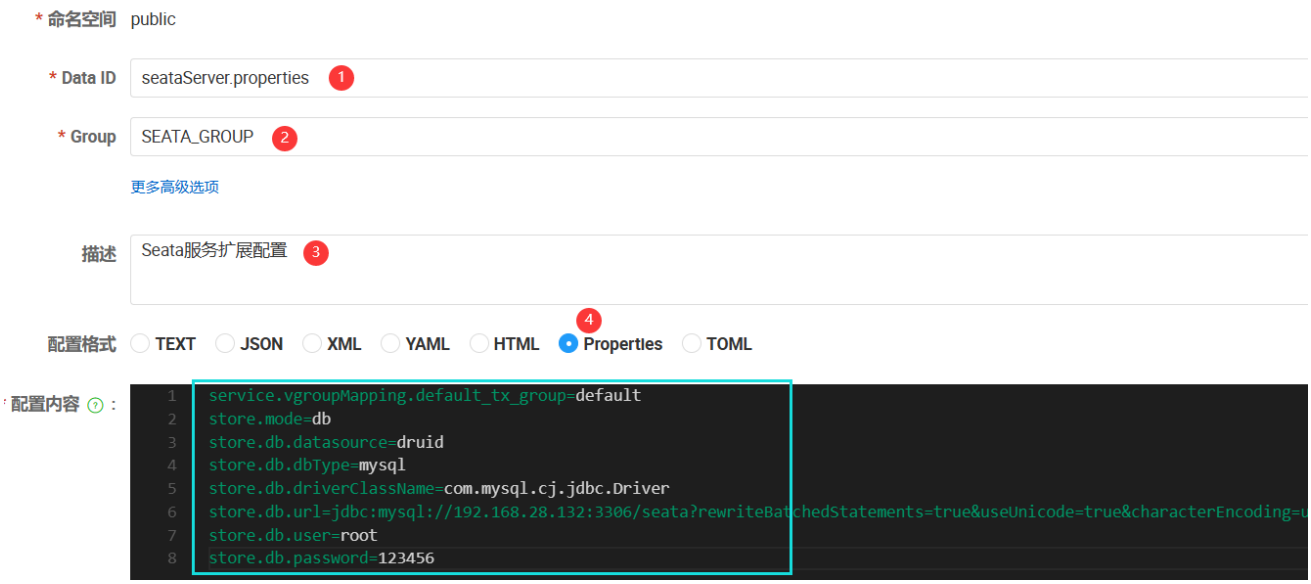

默认账号密码均为`seata`，需要修改密码的话在`application.yml`修改即可，下面是登录成功后的页面




# 3 扩展内容

## 3.1 配置中心填充

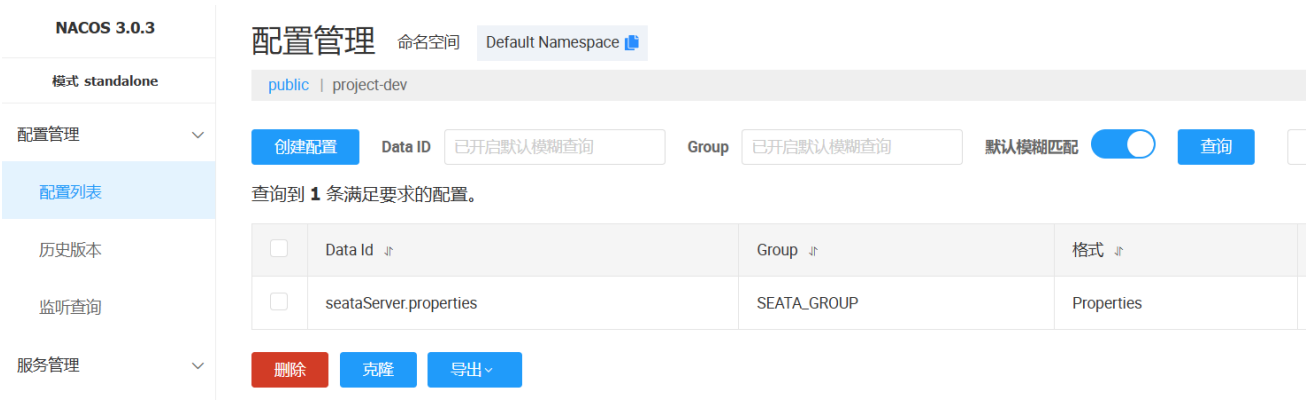
在你`seata`所在的`namespace`中新建配置，配置`data-id`为`seataServer.properties`，分组为`SEATA_GROUP`，这些属性和我们前面安装`Seata`修改的配置是匹配的，如下图所示。




配置内容示例

```
service.vgroupMapping.default_tx_group=default
# 存储模式使用数据库模式
store.mode=db
store.db.datasource=druid
store.db.dbType=mysql
store.db.driverClassName=com.mysql.cj.jdbc.Driver
store.db.url=jdbc:mysql://192.168.28.132:3306/seata?
rewriteBatchedStatements=true&useUnicode=true&characterEncoding=utf8&autoReconnec
t=true&useSSL=false&serverTimezone=Asia/Shanghai&allowPublicKeyRetrieval=true
store.db.user=root
store.db.password=123456
store.db.minConn=5
store.db.maxConn=30
store.db.globalTable=global_table
store.db.branchTable=branch_table
store.db.lockTable=lock_table
store.db.distributedLockTable=distributed_lock
store.db.vgroupTable=vgroup_table
store.db.queryLimit=100
store.db.maxWait=5000
store.db.druid.timeBetweenEvictionRunsMillis=120000
store.db.druid.minEvictableIdleTimeMillis=300000
store.db.druid.testWhileIdle=true
store.db.druid.testOnBorrow=false
store.db.druid.keepAlive=false
```

发布成功后，可以看到下图所示效果。




更多配置内容参考下面链接，选择性的复制配置到你的项目中，然后重启seata服务。

https://github.com/apache/incubator-seata/blob/2.x/script/config-center/config.txt

## 3.2 数据库表补充

需要在业务数据库中新增`undo_log`表，表的内容参考`资源/db/client-db.sql`脚本。